Olympiad Test Paper — Science (Class 7) — Paper 4

Chapter 19: Earth, Moon and the Sun

Paper 4 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	19 — Earth, Moon and the Sun
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: **+4** for each correct answer · **−1** for each wrong answer · **0** if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Reason from the actual Sun-Earth-Moon geometry, not from labels: phases come from the lit fraction we see (never Earth's shadow), eclipses need near-collinear alignment plus the right phase, time differences follow from 15 degrees of rotation per hour eastward, seasons follow from a fixed tilt not distance, and 'no atmosphere' is the single cause behind a whole family of lunar facts. Do not write on this paper — mark answers on your answer sheet.

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1. Two cities lie on the same line of latitude, but City P is 45 degrees of longitude east of City Q. When the clock reads 12:00 noon at City P, the local solar time at City Q is closest to:

(A) 3:00 in the afternoon
(B) 12:00 noon as well
(C) 9:00 in the morning
(D) 6:00 in the morning

2. A place on the equator is carried right around Earth's circle in 24 hours, while a place at high latitude traces a much smaller circle in the same 24 hours. Compared with the high-latitude place, a person standing on the equator is being carried eastward at:

(A) a faster speed, because the equator covers a longer circle in the same time
(B) a slower speed, because the equator is hotter
(C) exactly the same speed, since both take 24 hours
(D) no speed at all, because the equator faces the Sun

3. It is sunrise at a coastal town. Exactly six hours later, that same town has been carried by Earth's spin through roughly:

(A) a full turn, so it is sunrise again
(B) half a turn, so the Sun is setting
(C) no turn, so the Sun is still rising

(D) a quarter turn, so the Sun is now near its highest midday point

4. An airliner takes off heading due west and flies fast enough to keep the Sun at the same height in the sky for the whole trip. To an observer on the ground beneath, this means the plane is:

(A) moving west just fast enough to cancel out Earth's eastward spin, so it stays under the same patch of sunlight

(B) stopping the Sun from moving by its engines

(C) flying so fast that time goes backward

(D) matching the Moon's speed instead of the Sun's

5. When the clocks in London read 12:00 noon, India (about 75 degrees of longitude to the east) is well into the afternoon. Using 15 degrees per hour, India's solar time is ahead of London's by about:

(A) seven and a half hours | (B) five hours

(C) two hours | (D) twelve hours

6. Earth spins from west to east. A new artificial satellite is launched toward the east, in the same direction as the spin, rather than toward the west. The usual reason is that:

(A) the Sun pulls rockets only toward the east

(B) westward launches fall straight back down

(C) Earth's eastward spin already gives the rocket a head-start speed, so less fuel is needed

(D) the Moon blocks all western launch paths

7. A river in the Northern Hemisphere flows due south. Because of Earth's rotation, the flowing water tends to press a little harder against:

(A) its eastern bank, because water always turns left

(B) neither bank, because rivers are too slow to be affected

(C) the river bed straight down, not the banks

(D) its western bank, because moving things are deflected to the right in the Northern Hemisphere

8. Suppose Earth spun from east to west instead of its real west-to-east direction, with everything else unchanged. For an observer in India the Sun would then appear to:

(A) rise in the west and set in the east | (B) rise and set in the same place as now

(C) stop rising and setting altogether | (D) circle overhead without ever setting

9. A long-distance runner crosses a time-zone boundary while running due east and notices clocks jump forward by one hour. The most direct reason zones are set one hour apart is that:

(A) each 15-degree band of longitude meets the Sun one hour apart as Earth turns

(B) the Sun physically jumps one hour east at each boundary

(C) the Moon's phase advances one hour per zone

(D) Earth speeds up its spin at every border

10. On the same instant, it is mid-morning in India and the middle of the night in the United States. A single statement that explains both is:

(A) the Sun shines on India but is switched off over the United States

(B) the Moon's shadow blacks out the United States right then

(C) clouds cover the United States while India is clear

(D) the Sun lights only the half of Earth facing it, and these two places sit in the lit and unlit halves at that moment

11. On the June day when the North Pole leans most toward the Sun, a town in the Southern Hemisphere is experiencing:

(A) the height of its summer, with long warm days

(B) the depth of its winter, with short days and slanting rays

(C) exactly equal day and night, like an equinox

(D) no daylight at all for the whole season

12. Two beams of sunlight carry the same energy. One strikes the ground from straight overhead, the other comes in at a low slant. The slanting beam warms the ground less because it:

(A) loses energy on the longer path and arrives colder

(B) is a different colour with less heat in it

(C) is reflected away before it can warm anything

(D) spreads the same energy over a wider patch of ground, so each spot receives less

13. A student reasons: 'Earth is closest to the Sun in early January, so January must be the hottest month worldwide.' The flaw is that:

(A) Earth is actually farthest from the Sun in January

(B) distance has no effect on anything in space

(C) seasons are set by the axial tilt, not by the small change in distance, so the tilted-away hemisphere still has winter in January

(D) the Sun is dimmer in January than in July

14. Imagine Earth's axis were tilted at a much larger angle than its real 23.5 degrees, say close to 60 degrees, with everything else the same. The most direct effect would be:

(A) no seasons at all, because a bigger tilt cancels itself out

(B) a much longer year

(C) far more extreme seasons, because each hemisphere would lean much more sharply toward and away from the Sun

(D) the Sun would stop rising and setting

15. At the equinox in March, the day and night are nearly equal everywhere on Earth. This happens because at that point in the orbit:

(A) the North Pole leans most toward the Sun

(B) neither pole is leaning toward the Sun, so the dividing line between light and dark passes through both poles

(C) Earth is exactly halfway around in distance from the Sun

(D) the Moon shares its light equally over the planet

16. Earth takes about 365 and one-quarter days for one orbit. We handle the extra quarter-day by adding February 29 every four years. If instead we simply ignored those quarter-days forever, after many decades the calendar would drift so that:

(A) the year would suddenly become much shorter

(B) the dates would slowly slip out of step with the seasons, so eventually 'summer' months could fall in cold weather

(C) Earth would speed up its orbit to catch up(D) leap years would appear on their own

17. During the long polar summer, an observer at the North Pole sees the Sun circle around the sky without ever setting for weeks. The reason this place gets continuous daylight is that:

(A) Earth stops spinning over the pole in summer

(B) the Sun grows large enough to light the whole globe

(C) the tilt keeps the pole leaning toward the Sun, so Earth's spin never carries it into the dark half

(D) the Moon reflects extra sunlight onto the pole all night

18. A town near the equator finds its daytime length barely changes from June to December, while a town near the Arctic sees daylight swing from nearly all day to nearly all night across the year. The key reason for the difference is that:

(A) the equator is closer to the Sun all year

(B) the tilt changes how much of a high-latitude daily circle falls in the lit half far more than it changes an equatorial one

(C) high-latitude towns spin faster than equatorial ones

(D) the Moon blocks sunlight only at high latitudes

19. On the December day when the Southern Hemisphere leans most toward the Sun, the city of Sydney in Australia is having long, warm days. At that same time, the slanting weak rays and short days belong to:

(A) cities in the Northern Hemisphere, which are then leaning away from the Sun

(B) cities in the Southern Hemisphere along with Sydney

(C) every city on Earth equally (D) only cities exactly on the equator

20. A thin crescent Moon is seen low in the western sky shortly after sunset, with its bright curved edge facing down toward where the Sun has just set. The bright edge points that way because:

(A) the lit side always faces the Sun, which is just below the western horizon at that time

(B) the bright edge always points straight down regardless of the Sun

(C) Earth's shadow carves the dark side and forces the bright edge down

(D) the brighter side is simply the heavier side of the Moon

21. At first-quarter phase the Moon is a quarter of the way around its orbit since new Moon, and we see exactly half its disc lit. About a week later, at full Moon, the Moon is:

(A) halfway around its orbit, on the side of Earth opposite the Sun, so its whole near face is lit

(B) three-quarters around its orbit, between Earth and the Sun

(C) back at the start, between Earth and the Sun

(D) still a quarter around, having paused for a week

22. An observer notices the Moon is lit on its left side and growing thinner each night. Over the following week this Moon will:

(A) wax, growing toward full Moon

(B) continue to wane, shrinking toward new Moon

(C) stay exactly the same shape

(D) switch suddenly to a full Moon overnight

23. At full Moon the Moon rises in the east just as the Sun sets in the west; the two are on opposite sides of the sky. From this geometry, where will the full Moon be at local midnight?

(A) just rising on the eastern horizon

(B) high in the sky, near its highest point

(C) just setting on the western horizon

(D) below the horizon, not yet risen

24. A waxing crescent Moon sets in the west soon after the Sun, while a full Moon stays up all night. The reason a thin waxing crescent is visible for only a short time after sunset is that it:

(A) lies close to the Sun's direction in the sky, so it follows the Sun down soon after sunset

(B) is opposite the Sun, so it rises only at midnight

(C) makes its own light that fades within an hour

(D) is hidden inside Earth's shadow most of the night

25. Two students try to draw the Sun-Earth-Moon arrangement that produces a first-quarter Moon. The correct arrangement places the Moon so that the line from Earth to the Moon makes with the line from Earth to the Sun an angle of about:

(A) 0 degrees, with the Moon toward the Sun

(B) 180 degrees, with the Moon opposite the Sun

(C) 90 degrees, a right angle

(D) 45 degrees, halfway between

26. Half of the Moon is always lit by the Sun, yet from Earth we see the Moon change shape through the month. Both facts fit together because:

(A) the Moon really swells and shrinks each month

(B) we view that fixed lit half from changing angles, so the fraction of it we can see changes

(C) the Sun lights different total amounts of the Moon each night

(D) Earth's shadow trims the Moon to different shapes

27. The terminator is the line on the Moon dividing its lit part from its dark part. At a place on the Moon's surface that lies exactly on this line, the Sun is:

(A) directly overhead, at lunar noon

(B) right at the horizon, just rising or just setting there

(C) completely below the surface forever(D) replaced by Earth as the light source

28. A 'waxing gibbous' Moon shows more than half but not all of its disc lit and is growing toward full. In its orbit this puts the Moon:

(A) directly between Earth and the Sun (B) between new Moon and first quarter

(C) exactly at the full-Moon point already

(D) between the first-quarter and full positions, so we see most of the lit half

29. On a night with a thin crescent Moon, the rest of the Moon's dark disc can be faintly seen, softly glowing. This 'earthshine' is caused by:

(A) sunlight reflected off Earth onto the Moon's dark part and then back to us

(B) the Moon making a little of its own light in its dark areas

(C) heat escaping from inside the Moon (D) starlight collecting on the unlit side

30. An astronaut on the Moon watches Earth in the lunar sky and sees it go through phases too. When people on Earth see a full Moon, that same astronaut sees Earth as:

(A) a 'full Earth', fully lit (B) exactly half-lit, like a quarter phase

(C) completely invisible, because Earth makes no light

(D) a 'new Earth', mostly dark, because Earth's lit face is then turned toward the Sun and away from the Moon

31. A waning crescent and a waxing crescent both look like thin slivers but the lit edge is on opposite sides. Tracked over several nights, the way to tell which is which is that the:

(A) waxing crescent is warmer to the touch than the waning one

(B) waxing crescent makes its own light and the waning one reflects

(C) waxing crescent appears only in winter and the waning only in summer

(D) waxing crescent grows fatter each night while the waning crescent grows thinner

32. We always see essentially the same face of the Moon. A student claims this is because the Moon does not rotate at all. The clearest correction is that the Moon:

(A) truly never spins, which is why one face is fixed

(B) rotates exactly once for every orbit it makes, so the same side stays pointed at us

(C) spins far faster than it orbits, blurring the far side

(D) is held still by Earth's magnetism

33. Imagine the Moon kept orbiting Earth exactly as now but stopped spinning entirely. Over one full orbit, an Earth observer would then:

(A) get to see all of the Moon's surface, because the same face would no longer stay turned toward us

(B) still see only the same single face as now

(C) see the Moon vanish, since a still Moon cannot reflect light

(D) see the Moon flip over once each night

34. The far side of the Moon is sometimes wrongly called the 'dark side.' During the lunar day on the far side it is in fact:

(A) always in total darkness, never receiving sunlight

(B) lit only by light bounced from Earth

(C) brightly lit by the Sun, even though it can never be seen from Earth

(D) frozen forever because no sunlight reaches it

35. The Moon returns to the same point against the background stars in about 27.3 days, but the cycle of phases (full Moon to full Moon) takes about 29.5 days. The extra two days are needed because:

(A) the Moon pauses for two days at each full Moon

(B) Earth's shadow holds the Moon back each month

(C) while the Moon circles Earth, Earth has moved along its own orbit, so the Moon must travel a little farther to line up with the Sun again

(D) the Moon slows down near full and speeds up near new

36. One full day-night cycle on the Moon's surface, from one lunar sunrise to the next, lasts about a month rather than a day. This follows directly from the fact that the Moon:

(A) does not rotate, so the Sun never moves there

(B) turns once on its axis in about a month, so the Sun takes a month to come back around its sky

(C) spins once a day just like Earth (D) is lit by Earth instead of the Sun

37. A camera on a spacecraft can photograph the Moon's far side, but no one standing anywhere on Earth ever sees it directly. The reason is that the Moon's:

(A) far side is permanently dark and so gives no image to Earth

(B) far side is sealed inside Earth's shadow

(C) spin keeps the same face turned toward Earth, so the far side never rotates into our view

(D) far side faces away from the Sun and is therefore invisible

38. Astronauts' footprints from 1969 are expected to last for millions of years on the Moon. The single underlying reason behind their survival is that the Moon has:

(A) such strong gravity that the dust is pressed permanently flat

(B) no atmosphere and no liquid water, so there is no wind or rain to wear them away

(C) constant volcanoes that re-bake and harden the surface

(D) a thick air layer that seals the prints under pressure

39. Daytime temperatures on the Moon can climb above the boiling point of water while night temperatures plunge far below freezing. The single best explanation for this huge swing is that the Moon:

(A) sits much closer to the Sun than Earth does

(B) is made of materials that generate their own heat in sunlight

(C) has no atmosphere to trap and spread heat, so the surface heats and cools sharply

(D) spins so fast that one side bakes while the other freezes

40. At lunar noon an astronaut looks at the sky right next to the blazing Sun and finds it black, not blue. The reason the lunar daytime sky stays black is that the Moon has:

(A) no atmosphere to scatter sunlight, so apart from the Sun itself the sky stays dark

(B) a dark-coloured atmosphere that soaks up blue light

(C) moved into Earth's shadow at that moment(D) no sunlight reaching it at noon

41. On Earth sunset is followed by a gradual twilight, but on the Moon daylight gives way to darkness almost the instant the Sun dips below the horizon. The reason for the abrupt lunar change is that the Moon has:

(A) a thick atmosphere that blocks the Sun all at once

(B) such fast rotation that the Sun vanishes in seconds

(C) no atmosphere to scatter sunlight after the Sun sets, so there is no lingering glow

(D) its own light that it switches off at sunset

42. A meteoroid that would flare up as a bright 'shooting star' in Earth's sky instead strikes the Moon silently and gouges a crater. The reason it does not burn up before landing on the Moon is that the Moon has:

(A) gravity too weak to attract any meteoroids(B) a surface too cold to allow burning

(C) no atmosphere to heat it by friction, so it reaches the surface without burning

(D) a magnetic field that deflects the rock away

43. Astronauts dropped a hammer and a feather together on the Moon and they hit the ground at the same instant, unlike on Earth. This shows that on the Moon:

(A) there is no air, so no air resistance slows the lighter feather

(B) gravity is so strong it forces both to fall together

(C) there is zero gravity, so both simply float down

(D) the feather is heavier than it is on Earth

44. A solar eclipse can happen only at new Moon and a lunar eclipse only at full Moon. The reason both cannot happen at any phase is that an eclipse needs the Sun, Earth and Moon to be:

(A) set at right angles, as at the quarter phases

(B) very nearly in a straight line, which only the new and full positions provide

(C) at their greatest distances apart (D) spinning at the same rate

45. In a solar eclipse the Moon's shadow has a small dark central cone and a wider lighter surrounding cone. A person standing in the dark central cone (the umbra) sees:

(A) the Sun completely covered, a total eclipse(B) the Sun completely unaffected

(C) the Moon glowing coppery red (D) only the edge of the Moon clipping the Sun

46. During the same solar eclipse, an observer standing in the lighter outer part of the Moon's shadow (the penumbra) instead sees:

(A) the Sun fully blacked out, as in totality(B) no change to the Sun at all

(C) a total lunar eclipse instead

(D) only part of the Sun covered, a partial eclipse

47. The Sun is about 400 times wider than the Moon, yet during a total solar eclipse the Moon just covers the Sun's disc. This near-perfect fit is possible because the Moon is:

(A) actually almost as wide as the Sun in reality

(B) about 400 times closer to us than the Sun, so the two appear the same size

(C) made of denser material than the Sun(D) brighter than the Sun and so blots it out

48. During a total lunar eclipse the fully shadowed Moon often glows a dull coppery red rather than going black. The reason it is reddish is that Earth's atmosphere:

(A) bends some red sunlight around Earth's edge into the shadow, where it falls on the Moon

(B) produces its own red light shining onto the Moon

(C) sets the Moon's surface on fire

(D) reflects the red colour of Earth's oceans to the Moon

49. A learner expects a lunar eclipse at every full Moon, since Earth is then between the Sun and Moon. Lunar eclipses are far rarer than full Moons because:

(A) Earth's shadow is far too small to ever reach the Moon's orbit

(B) an eclipse actually needs a new Moon, not a full Moon

(C) the Moon's orbit is slightly tilted, so at most full Moons the Moon passes a little above or below Earth's shadow

(D) the Moon must make its own light for an eclipse

50. During a total solar eclipse only a narrow strip of Earth sees totality, while a much wider region sees only a partial eclipse. This pattern matches the shadow shape because the Moon's:

(A) umbra covers half the Earth while the penumbra is tiny

(B) shadow is the same darkness everywhere it falls

(C) shadow only ever produces partial eclipses

(D) small dark umbra touches only a narrow path while the wider penumbra spreads over a much larger area

51. A solar eclipse is seen only in daytime from places facing the Sun, while a lunar eclipse can be seen from the entire night side of Earth at once. The reason a lunar eclipse is visible so widely is that:

(A) the Moon makes its own light during an eclipse, visible from anywhere

(B) Earth's shadow lands on Earth's own surface for all to see

(C) the darkened Moon is simply seen by everyone for whom the Moon is above the horizon, the whole night half of Earth

(D) lunar eclipses change the daytime sky everywhere

52. Comparing the actual sizes of the Sun, Earth and the Moon, the correct order from largest to smallest is:

(A) Sun, then Moon, then Earth

(B) Earth, then Sun, then Moon

(C) Moon, then Earth, then Sun

(D) Sun, then Earth, then Moon

53. Listing the Moon, the Sun and the next-nearest star beyond the Sun in order of increasing distance from Earth (nearest first) gives:

(A) the Sun, then the Moon, then the next-nearest star

(B) the Moon, then the next-nearest star, then the Sun

(C) the next-nearest star, then the Sun, then the Moon

(D) the Moon, then the Sun, then the next-nearest star

54. The Sun looks like a big bright disc while other stars look like tiny twinkling points, even though some of those stars are far larger than the Sun. The reason is that the Sun is:

(A) the biggest star that exists, dwarfing all others

(B) the only real star, the rest being reflective rocks

(C) the only object in space that makes its own light

(D) vastly closer to Earth than any other star, so it looks large while they look like points

55. The Sun makes its huge output of light and heat by nuclear fusion of hydrogen into helium, while a planet like Jupiter, though large, shines only by reflected sunlight. The key difference is that:

(A) the Sun generates its own energy in its core, whereas the planet only bounces back sunlight

(B) the planet also runs on fusion but more weakly

(C) the Sun reflects light from other stars

(D) both make their own light, the Sun just more of it

56. Two high tides and two low tides occur at most coasts each day. Spring tides, the highest of all, occur at new Moon and full Moon because at those times the Sun and Moon:

(A) pull at right angles, partly cancelling each other

(B) stop pulling on the oceans entirely

(C) line up so their gravitational pulls on the oceans add together

(D) move to opposite sides of the galaxy

57. The weaker neap tides occur at the quarter Moons. At those phases the Sun and Moon are positioned so that, relative to Earth, they pull on the oceans:

(A) from exactly the same direction, adding up(B) not at all during those phases

(C) twice as strongly as at other times

(D) at roughly right angles, so their pulls partly cancel

58. Earth has one natural satellite while artificial satellites are built and launched by people. Both keep circling Earth instead of flying off into space because:

(A) the Sun pushes them back toward Earth with its light

(B) there is no force on them, so they drift in circles by themselves

(C) Earth's magnetism alone holds every satellite up

(D) Earth's gravity continuously pulls them toward it, bending their path into an orbit

59. The Moon is seen only because it reflects sunlight to the Earth. Suppose the Moon were shifted to twice its present distance from the Earth, with everything else unchanged. As seen from the Earth over a month, the Moon would:

(A) look exactly the same size but stop showing any phases at all

(B) look smaller and a little fainter, yet still pass through the same phases in the same order

(C) look larger, because a bigger share of its sunlit half would then face the Earth

(D) appear completely dark, because sunlight could no longer travel that far

60. A constellation such as Orion looks like a tidy group of stars, but its stars are actually:

(A) all the same distance from Earth, forming a real cluster

(B) physically touching one another in space(C) all members of our own Solar System

(D) at very different distances, only appearing to line up as seen from Earth

Solutions — Science (Class 7), Chapter 19: Earth, Moon and the Sun — Paper 4

Paper 4 — (progressively harder)

Marking: **+4** correct, **−1** wrong, **0** unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	B	21	A	31	D	41	C	51	C
2	A	12	D	22	B	32	B	42	C	52	D
3	D	13	C	23	B	33	A	43	A	53	D
4	A	14	C	24	A	34	C	44	B	54	D
5	B	15	B	25	C	35	C	45	A	55	A
6	C	16	B	26	B	36	B	46	D	56	C
7	D	17	C	27	B	37	C	47	B	57	D
8	A	18	B	28	D	38	B	48	A	58	D
9	A	19	A	29	A	39	C	49	C	59	B
10	D	20	A	30	D	40	A	50	D	60	D

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) Earth turns 15 degrees of longitude in one hour, so 45 degrees corresponds to three hours. A place farther west sees the Sun reach its noon position later, so when it is noon to the east it is still mid-morning, three hours earlier, to the west.

2. (A) Both complete one turn in the same 24 hours, but the equator's circle is the largest on Earth. Covering a longer path in equal time means a greater speed, so the equatorial observer moves fastest.

3. (D) A full turn is 24 hours, so six hours is one quarter of a rotation. Starting from sunrise on the eastern edge of the lit half, a quarter turn carries the town to face most directly toward the Sun, which is local midday.

4. (A) The Sun appears to slide west only because Earth turns east. If the plane flies west at the same rate that the ground turns east, the plane stays opposite the same point of the Sun, so the Sun seems frozen in the sky.

5. (B) Dividing the 75-degree difference by 15 degrees per hour gives five hours. Because India lies east, it meets the Sun earlier, so its clock runs ahead of London's by that amount.

6. (C) The launch site is already moving east with the turning Earth. Firing in that same direction adds the ground's speed to the rocket's, so the engines have less work to do to reach orbital speed.

7. (D) In the Northern Hemisphere Earth's spin deflects moving things to their right. For water travelling south, the right-hand side is the west, so the current crowds against the western bank.

8. (A) The Sun's apparent path is opposite to the spin. With the real eastward spin the Sun appears to move east-to-west; reverse the spin and the apparent motion reverses too, so the Sun would come up in the west.

9. (A) Earth turns 360 degrees in 24 hours, which is 15 degrees each hour. Splitting the globe into 15-degree bands therefore makes neighbouring bands experience the Sun's position one hour apart, so the standard clocks are stepped by an hour.

10. (D) Sunlight falls on only the half of the globe turned toward the Sun. India sits in the lit half and the United States in the unlit half at that instant; Earth's spin will later swap them.

11. (B) When the Northern Hemisphere tilts toward the Sun it is northern summer, so the Southern Hemisphere tilts away and has winter at the same time, with shorter days and weaker slanting sunlight.

12. (D) A slanting beam lands as a stretched-out oval covering more area than an overhead beam of the same width. The fixed energy is shared over more ground, so each square gets a smaller share and feels cooler.

13. (C) The slight distance change is far too small to drive the seasons. What matters is which hemisphere leans toward the Sun; in January the Northern Hemisphere leans away, so it has winter despite Earth being a touch closer.

14. (C) Seasons arise because the tilt changes how directly each hemisphere faces the Sun through the year. A larger tilt swings that angle through a bigger range, so summers and winters would be sharper and more extreme.

15. (B) Equal day and night occur when the axis leans neither toward nor away from the Sun. The terminator then runs through both poles, splitting every latitude's daily circle into equal lit and unlit halves.

16. (B) Each year the calendar would fall about a quarter-day behind Earth's true position in orbit. Decade after decade these slips accumulate, gradually shifting the calendar months away from the seasons they were meant to mark.

17. (C) While that pole leans sunward, it stays inside the lit half of Earth all the way around each daily turn. Rotation merely swings the Sun around the horizon, never dipping it below, so daylight is unbroken.

18. (B) The same axial tilt moves the dividing line of light and dark only slightly across an equatorial daily circle but dramatically across a near-polar one, so day length is nearly steady at the equator and wildly variable near the poles.

19. (A) When the Southern Hemisphere tilts toward the Sun it enjoys summer, so the Northern Hemisphere tilts away and has winter with short days and slanting rays at the very same moment.

20. (A) Whatever phase we see, the Moon's bright side is the half lit by the Sun. Just after sunset the Sun sits below the western horizon, so the crescent's lit edge bows toward that direction.

21. (A) From new Moon to first quarter is one quarter of the orbit; another quarter reaches full Moon at the halfway point. There the Moon lies opposite the Sun, so sunlight floods the entire face turned toward us.

22. (B) A Moon that is already thinning night after night is waning, heading from a past full Moon down toward new Moon. The lit sliver therefore keeps shrinking over the coming week.

23. (B) Because the full Moon sits opposite the Sun, it behaves like the Sun does at noon but at the opposite clock time. The Sun is highest at noon, so the Moon opposite it is highest at midnight.

24. (A) A young crescent is only a small angle from the Sun in the sky. As the Sun sinks below the western horizon, the nearby crescent follows shortly after, giving just a brief evening view.

25. (C) At first quarter we see exactly half the lit face, which happens when our line of sight to the Moon is at a right angle to the sunlight direction. So the Earth-to-Moon and Earth-to-Sun lines meet at about 90 degrees.

26. (B) The Sun always lights one whole half of the Moon. As the Moon orbits, our viewing angle on that lit half keeps changing, so we glimpse different fractions of it, which we read as the changing phases.

27. (B) The terminator marks where day meets night on the Moon. Standing on it, an observer would see the Sun grazing the horizon, exactly the moment of local sunrise or sunset there.

28. (D) Gibbous means more than half-lit. Growing toward full places the Moon past first quarter but not yet at full, so our line of sight catches most of its sunlit half.

29. (A) When the Moon shows us a thin crescent, Earth shows the Moon a nearly full, bright face. That earthlight bounces onto the Moon's night side and reflects back, dimly lighting the dark disc for us.

30. (D) At full Moon the Moon is opposite the Sun from Earth, so Earth's sunlit side faces the Sun and its night side faces the Moon. The astronaut therefore sees a dark 'new Earth' when we see a full Moon.

31. (D) Waxing means the lit part is increasing on its way toward full, and waning means it is decreasing toward new. Watching across nights, the growing sliver is waxing and the shrinking one is waning.

32. (B) If the Moon did not spin at all, it would show us every side over an orbit. Showing one face requires its spin to keep pace with its orbit, turning just enough each time to face us steadily.

33. (A) The same-face view depends on the Moon turning once per orbit. Stop that spin and the Moon would keep one direction fixed in space while orbiting, so different sides would face Earth and we would eventually see all of it.

34. (C) As the Moon rotates, every part of it including the far side faces the Sun in turn and is fully lit during its day. It is hidden from us by the same-face geometry, not by any lack of sunlight.

35. (C) The phase cycle depends on the Sun-Earth-Moon angle, not on the stars alone. Since Earth shifts along its orbit during the month, the Moon needs roughly two extra days of travel to catch the same alignment with the Sun.

36. (B) How long a day lasts on a body depends on how fast it spins. The Moon takes about a month to rotate once, so the Sun crosses its sky and returns over roughly that same month-long stretch.

37. (C) Because the Moon turns once per orbit, the same hemisphere always points at Earth. The far side does get sunlight and can be photographed from space, but it never swings around to face our planet.

38. (B) On Earth wind and water erode marks quickly. The Moon's lack of an atmosphere and of liquid water means there is essentially nothing to disturb the dust, so the prints stay almost unchanged for ages.

39. (C) An atmosphere acts like a blanket, holding and sharing heat. With no air at all, the lit ground overheats and the dark ground loses its heat to space with nothing to even out the temperatures.

40. (A) Earth's sky is blue because air scatters sunlight in all directions. The Moon has no air to do this scattering, so away from the direct disc of the Sun the sky has no glow and appears black even by day.

41. (C) Twilight on Earth comes from air still scattering sunlight after the Sun has set. With no atmosphere, the Moon has nothing to keep the sky lit once the Sun is down, so darkness arrives almost at once.

42. (C) A 'shooting star' glows because friction with air heats the meteoroid. The Moon has no air to provide that friction, so the rock arrives at full speed and slams into the surface, forming a crater.

43. (A) On Earth air resistance holds back a light feather more than a hammer. With no atmosphere on the Moon, both fall under gravity alone and so reach the ground together, even though the Moon's gravity is weaker than Earth's.

44. (B) Casting one body's shadow onto another requires near-collinear alignment. The Moon lies between Earth and Sun only at new Moon, and Earth lies between them only at full Moon, so eclipses are pinned to those two phases.

45. (A) The umbra is the full-shadow region where the Moon blocks the entire Sun. An observer inside it experiences totality, with the Sun's bright disc hidden behind the Moon.

46. (D) The penumbra is the partial-shadow region where the Moon hides only part of the Sun. Anyone there sees a bite taken out of the Sun rather than complete coverage.

47. (B) A small near object can look as big as a huge far one if the distance offsets the size. The Moon is roughly 400 times nearer than the Sun, so its 400-times-smaller width fills the same angle in the sky.

48. (A) Earth's air bends and filters sunlight passing its edge, letting mostly red light slip into the shadow. That faint red light reaches the eclipsed Moon, giving it a coppery glow instead of total blackness.

49. (C) The Moon's path is tipped relative to Earth's orbit, so usually at full Moon the Moon slides just above or below Earth's shadow cone. Only when a full Moon falls near a crossing point does the Moon dive into the shadow and eclipse.

50. (D) The umbra, where the Sun is fully blocked, is small and traces a thin track across Earth. The surrounding penumbra, where the Sun is only partly blocked, is broad, so far more people see a partial than a total eclipse.

51. (C) A lunar eclipse darkens the Moon itself, so anyone who can see the Moon, that is the entire night side, witnesses it. A solar eclipse instead needs you to be in the Moon's small shadow track on Earth, which is far more limited.

52. (D) The Sun is by far the largest of the three. Earth is much bigger than the Moon, so the ranking from largest to smallest is the Sun, then Earth, then the Moon.

53. (D) The Moon is our closest neighbour, the Sun is much farther but still the nearest star, and any other star lies enormously farther still. So nearest first is the Moon, the Sun, then the next-nearest star.

54. (D) Apparent size in the sky depends on distance. The Sun is overwhelmingly nearer than other stars, so it spreads into a disc while even larger but far more distant stars shrink to mere points of light.

55. (A) A star such as the Sun powers itself by fusion deep inside, producing its own light. A planet has no such furnace and is visible only because it reflects light it receives from the Sun.

56. (C) Tides are driven by the combined pull of Moon and Sun. At new and full Moon the three bodies line up, so the two pulls reinforce one another, producing the extra-large spring tides.

57. (D) At a quarter Moon the Sun-Earth-Moon angle is about a right angle. The two pulls then work across each other and partly cancel, giving the smaller tidal range we call neap tides.

58. (D) A satellite would travel in a straight line if nothing pulled on it. Earth's gravity tugs it constantly toward Earth, curving that straight-line motion into a closed orbit so it neither escapes nor falls in.

59. (B) The cycle of phases is set by the changing relative positions of the Sun, Earth and Moon, not by how far the Moon is from the Earth, so the new -> crescent -> first quarter -> full -> waning sequence would still run in the same order. A more distant Moon covers a smaller angle in the sky, so it looks smaller, and its reflected light is spread over a greater distance, so it looks a little fainter. Moving it farther does not change which fraction of its lit half points toward the Earth at a given configuration, so it would not look larger; and sunlight easily travels far past the Moon (it lights the whole Solar System), so the Moon would not go dark.

60. (D) A constellation is just a chance pattern in our line of sight. Its stars usually lie at hugely different distances and are not truly close together; they only seem grouped from our particular viewpoint.